

Experiment 9

Question: Experiment to understand Matplotlib library use cases in Data Science through visualization.

Aim

To understand the use cases of the Matplotlib library in data science by creating different types of visualizations.

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

def exp_9():
    rng = np.random.default_rng(1)
    months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
    sales = [250, 300, 280, 350, 400, 380]
    hours = rng.uniform(1, 10, 40)
    marks = 35 + 6 * hours + rng.normal(0, 6, 40)
    ages = rng.normal(35, 9, 500)
    groups = [rng.normal(m, 8, 100) for m in (50, 60, 70)]
    shares = [40, 30, 20, 10]

    print("Plot types demonstrated and the data science task each supports")
    print("1. Line plot      -> trend over time")
    print("2. Bar chart       -> compare categories")
    print("3. Scatter plot      -> relationship between two variables")
    print("4. Histogram         -> distribution of one variable")
    print("5. Box plot          -> spread and outliers across groups")
    print("6. Pie chart         -> share of each part in a whole")
    r = np.corrcoef(hours, marks)[0, 1]
    print(f"\nCorrelation between study hours and marks = {r:.3f}")
    print(f"Mean age = {ages.mean():.2f}, Std = {ages.std():.2f}")

    fig, ax = plt.subplots(2, 3, figsize=(14, 8))
    ax[0, 0].plot(months, sales, marker="o", color="green")
    ax[0, 0].set_title("Line plot: Monthly Sales")
    ax[0, 1].bar(months, sales, color="steelblue")
    ax[0, 1].set_title("Bar chart: Sales by Month")
    ax[0, 2].scatter(hours, marks, color="crimson")
    ax[0, 2].set(title="Scatter: Hours vs Marks", xlabel="Hours", ylabel="Marks")
    ax[1, 0].hist(ages, bins=20, color="orange", edgecolor="black")
    ax[1, 0].set_title("Histogram: Age Distribution")
    ax[1, 1].boxplot(groups)
    ax[1, 1].set_xticklabels(["A", "B", "C"])
    ax[1, 1].set_title("Box plot: Scores by Group")
    ax[1, 2].pie(shares, labels=["Python", "Java", "C++", "R"], autopct="%1.0f%%")
    ax[1, 2].set_title("Pie chart: Language Share")
    plt.suptitle("Matplotlib Use Cases in Data Science", fontsize=14)
    plt.tight_layout()
    plt.show()
```

exp_9()

Output

Plot types demonstrated and the data science task each supports

1. Line plot -> trend over time
2. Bar chart -> compare categories
3. Scatter plot -> relationship between two variables
4. Histogram -> distribution of one variable
5. Box plot -> spread and outliers across groups
6. Pie chart -> share of each part in a whole

Correlation between study hours and marks = 0.958

Mean age = 34.81, Std = 8.57

Graph

Matplotlib Use Cases in Data Science

